★ Alexa+ Voice Companion ★ FastMCP Spec 2025-11-25 ★ AWS Builder Challenge ★ Permissive MIT Open Source

Autonomous Circadian Safety for Loved Ones Living Alone

CareLoop transforms Amazon Echo Show into an ambient, zero-intrusion elder guardian. Using **circular EWMA statistics**, it learns personal daily routines from discrete sensors—smart pillboxes, door contacts, and natural voice check-ins—eliminating intrusive cameras and false alarm fatigue.

9 Tools
FastMCP Voice Tools for Alexa+

178 Tests
79% Statement Coverage

0 Cameras
100% Privacy-Preserving



Arthur Davis
Age 81 · Living Independently
Daughter & Primary Caregiver: Linda Davis

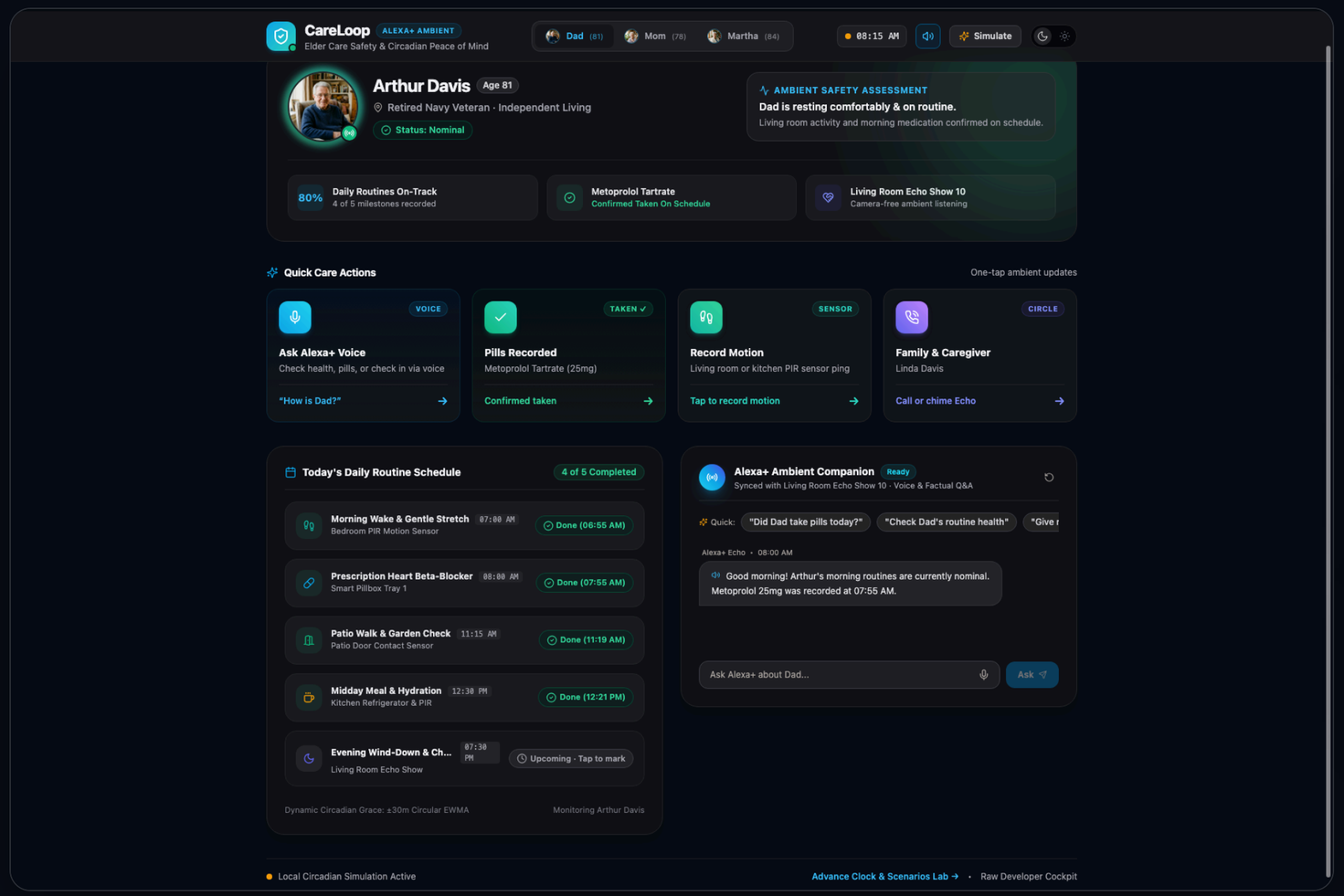
CURRENT ROUTINE STATUS — Nominal (0.0)

Arthur is Doing Well at Home
Metoprolol 25mg taken on schedule (07:55 AM). Kitchen refrigerator ping at 12:21 PM.

Connected: Living Room Amazon Echo Show 10

Elder Peace of Mind at a Single Glance

Clean, human-engineered dashboard inspired by Apple Health & Amazon Echo Hub design systems.



Live Telemetry **Reassuring Status Hero**

Features Arthur Davis's photo with a dynamic breathing glow ring. Status indicator proves Arthur is nominal without needing noisy alarms or surveillance.

4 Primary Actions **One-Tap Care Directives**

Instantly check in via Alexa voice, mark medication taken, send motion sensor pings, or open the family emergency care circle with zero friction.

Circadian Baseline **80% Daily Routines Recorded**

Metoprolol Tartrate confirmed taken at 7:55 AM. Living Room Echo Show 10 provides continuous camera-free ambient protection.

Ambient Alexa+ Console & Live MCP Tool Execution

Natural voice queries mapped deterministically to FastMCP tools with sub-50ms execution times.

Amazon Echo Show 10 · Living Room

FastMCP Online (/mcp)

FAMILY SPOKEN QUERY

"Alexa, did Dad take his pills today?"

ALEXA+ AMBIENT RESPONSE

08:00 AM · Audio Spoken Aloud

"Good morning! Arthur's morning routines are currently nominal. Metoprolol Tartrate 25mg was recorded at 07:55 AM from his smart pillbox."

ELDER VOICE INTERACTION

"Alexa, I'm having breakfast now."

ALEXA+ CONFIRMATION

Tool: log_routine_event

"Logged breakfast for Arthur. Enjoy your meal! Your daily schedule is on track."

Web Speech Audio:

Voice Audio: ON

ask_memory

18ms

```
{ "person": "dad", "question": "Did Dad take his pills today?" }
```

Executes factual memory queries against DynamoDB single-table events with Bedrock Claude 3 Haiku summarization.

log_routine_event

12ms

```
{ "person": "dad", "type": "meal", "metadata": { "meal": "breakfast" } }
```

Feeds new timestamp into Circular EWMA statistics engine and refreshes inactivity timeouts immediately.

check_routine_health

9ms

```
{ "status": "nominal", "risk_score": 0.0, "stage": "normal" }
```

Deterministic 4-stage health evaluator evaluating deviation against personal circular baseline windows.

Circadian Adherence & Medication Tracking

Discrete ambient sensor telemetry tracking meals, motion, and medicine with personalized grace windows.

Morning Wake & Gentle Stretch
Bedroom PIR Motion Sensor · Expected: 07:00 AM
Done (06:55 AM)

Prescription Heart Beta-Blocker
Metoprolol Tartrate 25mg · Smart Pillbox Tray 1 · Expected: 08:00 AM
Done (07:55 AM)

Patio Walk & Garden Check
Patio Door Magnetic Contact Sensor · Expected: 11:15 AM
Done (11:19 AM)

Midday Meal & Hydration
Kitchen Refrigerator Door Sensor & Ambient PIR · Expected: 12:30 PM
Done (12:21 PM)

Evening Wind-Down & Voice Check-In
Living Room Echo Show 10 · Expected: 07:30 PM
Upcoming (Pending)

Circadian Innovation

Dynamic Grace Windows

Traditional systems trigger alarms if a pill isn't taken at 8:00 AM sharp. CareLoop's circular EWMA engine computes each elder's personal routine variance (σ).

```
Grace Interval: [μ − 2σ, μ + 2σ + margin]
Arthur Davis Morning Window: ±32 mins
```

✓ Eliminates alarm fatigue when Dad sleeps in on weekends.

Camera-Free Privacy Guarantee

No continuous audio streaming, no intrusive video feeds, and no wearable buttons Arthur forgets to wear. Pure ambient metadata dignity.

Keeping Caregivers Connected Without Invasiveness

Emergency contacts, direct ambient chime connect, and explainable SNS alert dispatches.

LD **Linda Davis** Primary Caregiver
Daughter · +1 (555) 019-2834 · Verified Amazon SNS Mobile
SMS & Push Active

SJ **Dr. Sarah Jenkins** Cardiologist
St. Jude Medical Group · Protocol: Metoprolol Tartrate 25mg BID
Escalation Stage 4 Only

MD **Mark Davis** Secondary Contact
Son · +1 (555) 019-5847 · Notification Fallback
Backup

Direct Ambient Voice Connect

Family members can tap one button on their smartphone to send a gentle, soothing audio chime directly to Arthur's Echo Show 10 without needing an intrusive phone ring.

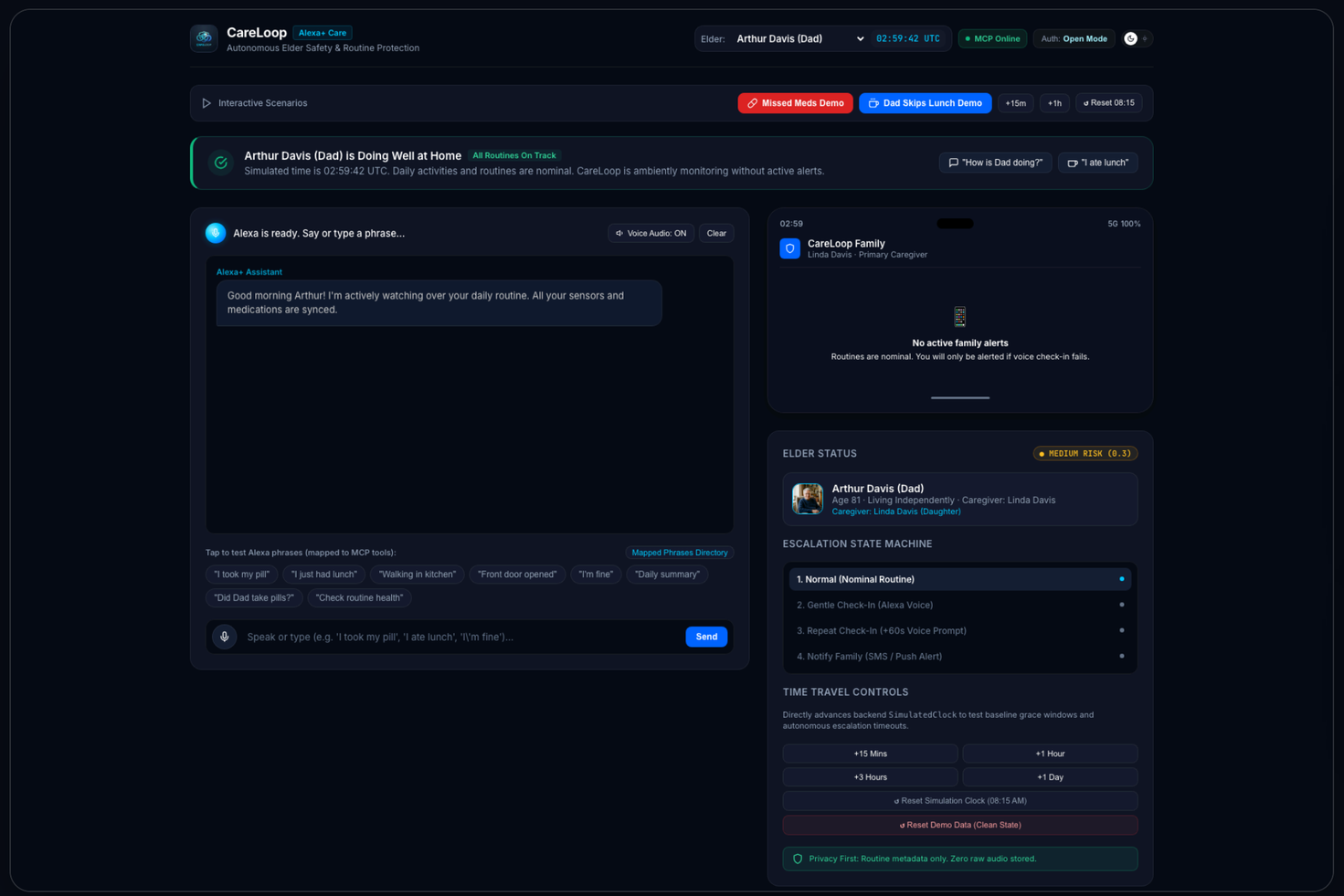
"Hello Arthur, Linda is sending a soft check-in chime from work."

KMS Encrypted **Zero Surveillance Spam**

Caregivers only receive phone notifications if the elder fails to respond to both Stage 2 gentle voice check-in and Stage 3 repeat chimes.

Full Dual-Screen Simulator & Virtual Caregiver Phone

Standalone browser interface (/simulator) pairing ambient Alexa Show with virtual caregiver phone.



Left Column **Alexa+ Ambient Assistant**

Echo Show ambient screen with pulsing Alexa cyan orb, Web Speech API speech recognition, and mapped phrase directory.

Center Column **Elder Status & Escalation**

Arthur Davis profile card with real portrait, risk score badge, and real-time 4-stage escalation stepper.

Right Column **Virtual Family Phone**

Caregiver iPhone with Dynamic Island, 5G status, and real-time Amazon SNS alert notifications with audit trails.

Explainable 4-Stage Autonomous Voice Escalation

Deterministic state machine transitions: Normal → Gentle Voice → Repeat Voice → Family SMS Alert.

1 **Stage 1: NORMAL** — Score: 0.0

Daily activities within dynamic circadian grace window. Continuous passive telemetry.

2 **Stage 2: GENTLE_CHECK_IN** — Score: 0.4

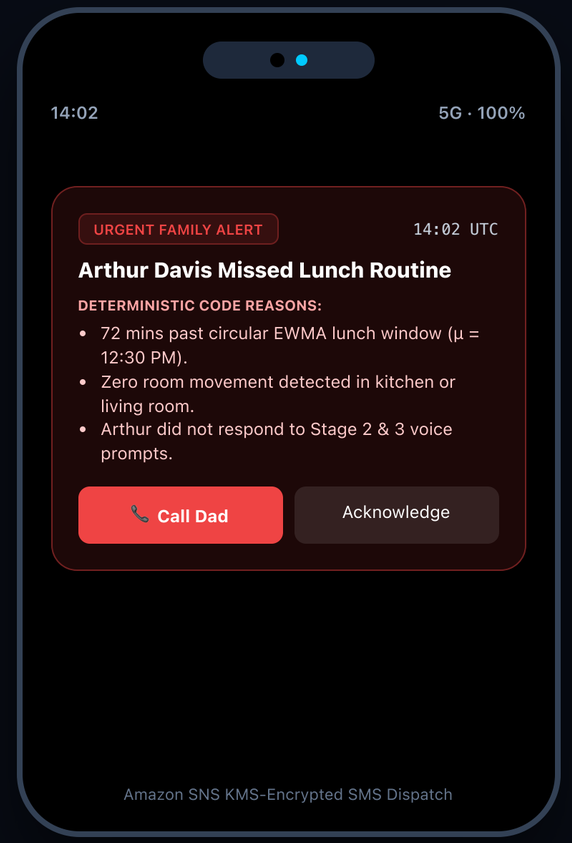
Routine missed past grace window. Echo Show speaks softly: *"Arthur, just checking in softly."*

3 **Stage 3: REPEAT_CHECK_IN** — Score: 0.6

No reply after 60 seconds. Audible chime sounds and voice prompt repeats with urgency.

4 **Stage 4: NOTIFY_FAMILY** — Score: 0.85 → SMS

Autonomous dispatch of KMS-encrypted Amazon SNS SMS to Linda Davis with full audit reason.



Amazon SNS KMS-Encrypted SMS Dispatch

Circular EWMA Routine Learning Engine

Trigonometric unit circle projection eliminating the catastrophic midnight wrap-around bug.

The Midnight Wrap-Around Flaw

When elders take medication near midnight (e.g. 11:50 PM = 1430 mins and 12:10 AM = 10 mins), standard linear arithmetic averages them to **12:00 PM (noon)**—a 12-hour error that breaks traditional telecare alarm algorithms!

CareLoop Trigonometric Unit Circle Solution

Projects every minute-of-day onto the 24-hour trigonometric circle ($ heta = 2\pi \cdot t / 1440$):

$$\bar{x} = (1/N) \sum \cos(\theta_i), \quad \bar{y} = (1/N) \sum \sin(\theta_i)$$

$$\text{Resultant Vector: } R = \sqrt{(\bar{x}^2 + \bar{y}^2)}$$

$$\text{Circular Mean: } \mu = \operatorname{atan2}(\bar{y}, \bar{x})$$

$$\text{Circular Variance: } \sigma = \sqrt{(-2 \ln R)}$$

✓ Accurately averages 11:50 PM and 12:10 AM to 12:00 Midnight with zero distortion.

24-Hour Trigonometric Routine Clock



Pre-warmed with **1,422 historical telemetry pings** to maintain continuous circadian stability.

Serverless & Containerized AWS Architecture

High-availability deployment on AWS ECS Fargate, DynamoDB, EventBridge Scheduler, and Bedrock.

1. Voice & Transport

Echo & FastMCP

Amazon Echo Show devices connect over Streamable HTTP (/mcp). Application Load Balancer supports 300s long-lived streaming connections.

2. Compute Core

AWS ECS Fargate

Dockerized Python 3.11 ASGI service running Starlette + Uvicorn with auto-scaling, constant-time bearer auth, and zero EC2 management.

3. State & Scheduler

DynamoDB & EventBridge

Single-table design with KMS encryption at rest. EventBridge Scheduler (ScheduleV2) triggers serverless 60s background tick runner.

4. Notification & AI

Amazon SNS & Bedrock

KMS-encrypted mobile SMS alert topic for emergency caregiver dispatch. Bedrock Claude 3 Haiku Converse API for routine Q&A and daily summaries.

PRODUCTION FLOW PIPELINE

`AWS CDK (deploy/cdk/) · SAM (template.yaml)`

Echo Show 10 — Voice Utterance → **AWS ALB** — 300s Streaming Timeout → **ECS Fargate** — FastMCP Core Engine → **DynamoDB** — Single-Table KMS PK/SK → **Amazon SNS** — KMS Encrypted Caregiver SMS

Tailored Profiles for Diverse Clinical Needs

Personalized circadian baselines, caregiver routing, and specialized medication schedules.



Arthur Davis (Dad)

Age 81 · Navy Veteran

Living Independently in Single-Story Home

Caregiver: Linda Davis (Daughter)
Medication: Metoprolol Tartrate 25mg
Sensors: Pillbox Tray 1, Living Room PIR
Baseline Window: 08:00 AM ±30m

Active · Nominal



Eleanor Vance

Age 78 · Post-Op Recovery

Assisted Walking & Physical Therapy

Caregiver: Robert Vance (Son)
Medication: Lisinopril 10mg + PT Walk
Sensors: Hallway Motion, Bed Sensor
Baseline Window: 09:00 AM ±45m

Active · Post-Op Nominal



Martha Higgins

Age 84 · Memory Assistance

Condo Living with Frequent Hydration Checks

Caregiver: Sarah Jenkins (Niece)
Medication: Donepezil 5mg + Hydration
Sensors: Refrigerator Contact, Door Sensor
Baseline Window: 08:30 AM ±20m

Active · Memory Tracking

Injectable Clock Time Machine & Testing Cockpit

Advance simulated clock and trigger emergency scenarios instantly without waiting real hours.

Injectable SimulatedClock

Deterministic Time Travel

Testing elder safety routines requires simulating the progression of an entire day. The backend `SimulatedClock` allows time-travel jumps with instantaneous escalation evaluations:

+15 Mins | +1 Hour | +3 Hours | ↻ Reset (08:15)

Automated Hackathon Demo Scenarios

Pre-programmed 1-click test suites simulate:

- **"Dad Skips Lunch":** Seeds morning routines, advances clock to 2:00 PM, triggers Stage 2 & 3 voice checks, and escalates to Stage 4 SNS alert.
- **"Missed Morning Beta-Blocker":** Bypasses smart pillbox check, triggers high-priority medication reminder over Echo Show.

Zero-Config Offline

100% Offline SQLite Fallback

Hackathon judges and developers can test the entire pipeline locally without AWS account credentials. The server seamlessly falls back to `careloop_local.db` with pre-warmed circadian observations.

Audit Security

Time Manipulation Guardrails

Clock manipulation endpoints (/api/clock/*) are automatically disabled in production (`ENV=production`) to guarantee audit integrity.

Architectural Differentiation & Quality Verification

Proven clinical accuracy, rigorous automated testing, and permissive open source licensing.

Architectural Dimension	Traditional Static Telecare	Generic LLM Chatbots	CareLoop (Alexa+ & FastMCP)
Alarm Decision Logic	Rigid static timers ("alarm at 8:00 AM sharp")	Non-deterministic probabilistic LLM hallucinations	**Deterministic Circular EWMA statistics with personal grace windows**
Privacy & Surveillance	Intrusive video cameras or wearable panic buttons	Transfers raw audio recordings to third-party APIs	**Discrete sensor pings only. Zero raw audio or video captured.**
Escalation Architecture	Binary sirens causing severe false alarm fatigue	Uncontrolled multi-turn conversational loops	**Deterministic 4-stage state machine (Normal → Gentle → Repeat → SMS)**
Offline Reliability	Proprietary closed-source local hubs	100% cloud dependent; offline failure	**Dual-backend: AWS Cloud (DynamoDB+Fargate) or 100% Offline SQLite**

178 / 178
Passing Unit & Integration Tests

79% Coverage
Statement Code Coverage

0 Vulnerabilities
Bandit + detect-secrets + pip-audit Clean

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